

JEE ADVANCED - 9 | PAPER - 2

JEE 2022

Date 27th February, 2022	Maximum Marks 204	Duration 3 Hours
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General Instructions

1. The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **Three** sections (Section 1, Section 2 & Section 3).
2. **Section 1** contains **8 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
3. **Section 2** contains **4 Paragraphs containing 2 Questions** on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 3** contains **4 Column-Match sets**. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
5. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION – 1

- This section consists of Eight **(08)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+4 If only (all) the correct option(s) is(are) chosen
Partial Marks:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks:	+1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	–1 In all other cases.

SECTION – 2

- This section has Four **(04)** Paragraphs containing **TWO (02) Questions** on each paragraph. Each question has **FOUR** options. **ONLY ONE** of these four option is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct option is chosen
Zero Marks:	0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	–1 In all other cases.

SECTION – 3

- This section has Four **(04)** Column-Match sets. Each set contains statements given in two columns. Statements in column I have to be matched with statements in column II. One or more than one statements in column II can be matched with a statement in column I.
- Each set has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the option corresponding to the correct combination is chosen
Zero Marks:	0 If none of the options is chosen (i.e., the question is unanswered)
Negative Marks:	–1 In all other cases.

PART I : PHYSICS

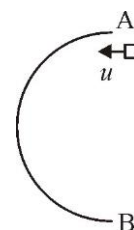
68 MARKS

SECTION-1

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

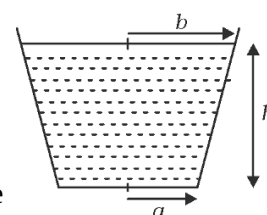
1. A semicircular ring of radius R is fixed on a smooth horizontal table. A small block is projected with a speed u so as to enter the ring at end A. Initial velocity of the block is along tangent to the ring at A and it moves on the table remaining in contact with the inner wall of the ring. The coefficient of friction between the block and the ring is μ .

- (A) Time after which the block will exit the ring at B is $t = \frac{R}{\mu u} e^{\pi\mu}$
- (B) Time after which the block will exit the ring at B is $t = \frac{R}{\mu u} [e^{\pi\mu} - 1]$
- (C) Speed of the block when it leaves the ring at A is $v = u(1 - e^{-\pi\mu})$
- (D) Speed of the block when it leaves the ring at A is $v = ue^{-\pi\mu}$



2. A bucket in the shape of the frustum of a cone of radii a and b and height h is filled completely with water of density ρ . Neglect the effects of atmospheric pressure. Which of these statements are correct?

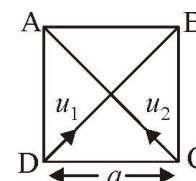
- (A) The pressure force on the flat circular base of the bucket is $\rho\pi gha^2$
- (B) The pressure force on the flat circular base of the bucket is equal to the weight of the water in the bucket
- (C) The pressure force on the curved surface of the bucket is $\rho\pi gh(b^2 - a^2)$
- (D) The net force on the bucket due to the water is equal to the weight of the water in the bucket



3. Two balls start to move along the diagonal DB and CA of a square of side ' a '. The balls reflect to follow the same path every time they reach a vertex. Speeds of balls are u_1 and u_2 respectively.

Which of the following statements is/are true for the above situation?

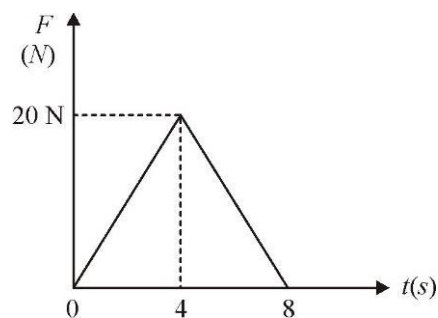
- (A) Balls always collide irrespective of their velocities.
- (B) Balls do not collide if ratio of velocities of balls (C) to ball (D) is EVEN.
- (C) If balls collide, ratio of distance traveled by both balls is same as the ratio of magnitudes of velocities irrespective of their magnitudes of velocities.
- (D) If balls do collide, ratio of total displacement of balls from start to the time of collision is always 1.



4. A sound wave of frequency f travels horizontally to the right. It is reflected from a large vertical plane surface moving to left with a speed v . The speed of sound in medium is c . Choose the correct options.

- (A) The number of waves striking the surface per second is $f \frac{(c+v)}{c}$
- (B) The wavelength of reflected wave is $\frac{c(c-v)}{f(c+v)}$
- (C) The frequency of the reflected wave received by a stationary observer is $f \frac{(c+v)}{(c-v)}$
- (D) The number of beats heard by stationary listener to the left of the reflecting surface is $\frac{vf}{c-v}$

5. A box of mass 4kg placed on a horizontal floor, is acted upon by a horizontal force F that varies with time as shown in the given graph. If coefficients of static and kinetic frictions are 0.25, on which of following conclusions can you arrive at? (Take $g = 10 \text{ ms}^{-2}$)
- (A) The box starts moving at the instant $t = 2.0 \text{ s}$
- (B) The maximum velocity acquired by the box is 5.0 m/s
- (C) The box stops at the instant $t = 9.0 \text{ s}$
- (D) None of these



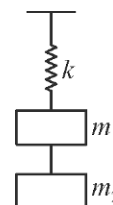
6. A particle of charge q and mass m moves rectilinearly under the action of an electric field $E = \alpha - \beta x$. Here, α and β are positive constants and x is the distance from the point where the particle was initially at rest. Then:
- (A) the motion of the particle is oscillatory
- (B) the amplitude of the particle is $\frac{\alpha}{\beta}$
- (C) the mean position of the particle is at $x = \frac{\alpha}{\beta}$
- (D) the maximum acceleration of the particle is $\frac{q\alpha}{m}$

7. The root mean square velocity of the molecules of a monoatomic gas at 300 K and pressure 10^5 Pa is $250\sqrt{3} \text{ m/s}$. Take $R = \frac{25}{3} \text{ J/mol K}$. Choose correct options.

- (A) Density of the gas at these conditions is 1.6 kg/m^3
- (B) Density of the gas at these conditions is 1.2 kg/m^3
- (C) Mass of 1 mole of this gas is 40 g
- (D) Kinetic energy of molecules of 0.2 m^3 of this gas at these conditions is 30 kJ

8. Two blocks of mass m_1 and m_2 are suspended by a spring of constant k as shown in the figure. Initially, the blocks are in equilibrium. Now the massless string connecting the blocks is cut. Which of these options are correct ?

- (A) Net force acting on m_1 just after string is cut is zero
- (B) m_1 starts oscillating with amplitude $\frac{m_2 g}{k}$
- (C) m_1 starts oscillating with frequency $\frac{1}{2\pi} \sqrt{\frac{k}{m_1}}$
- (D) Maximum extension in spring after string is cut is $\frac{(m_1 + 2m_2)g}{k}$



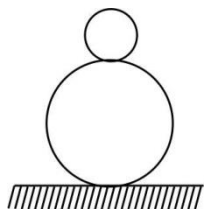
SECTION-2

This section consists of **FOUR (04) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question 9 - 10

A small ball (of negligible size) is placed over a sphere of same mass and radius 1 m as shown. All surfaces are smooth. A slight push is given to the ball. When the ball makes an angle of 30° with the vertical, speed of sphere is v . ($g = 10 \text{ m/s}^2$)

9. Value of v in m/s is :



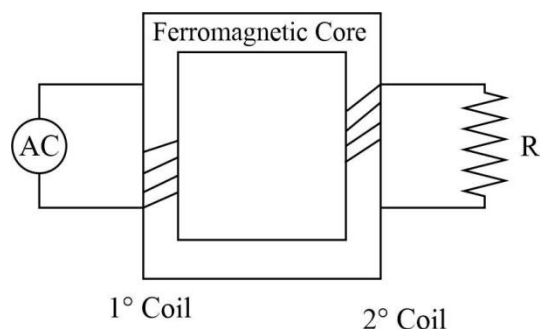
- (A) 0.9 (B) 1.6 (C) 2.4 (D) 0.4

10. If mass of ball and sphere be 1 kg each then the normal reaction between the two at this instant is :

- (A) 5.62N (B) 7.68N (C) 4.81N (D) 3.47N

Paragraph for Question 11 - 12

A physics lab is designed to study the transfer of electrical energy from one circuit to another by means of a magnetic field using simple transformers. Each transformer has two coils of wire electrically insulated from each other but wound around a common core of ferromagnetic material. The two wires are close together but do not touch each other.



The primary (1°) coil is connected to a source of alternating (AC) current. The secondary (2°) coil is connected to a resistor such as a light bulb. The AC source produces an oscillating voltage and current in the primary coil that produces an oscillating magnetic field in the core material. This in turn induces an oscillating voltage and AC current in the secondary coil.

Students collected the following data comparing the number of turns per coil (N), the voltage (V) and the current (I) in the coils of three transformers

	Primary Coil			Secondary Coil		
	N_{1°	V_{1°	I_{1°	N_{2°	V_{2°	I_{2°
Transformer-1	100	10V	10A	200	20V	5A
Transformer-2	100	10V	10A	50	5V	20A
Transformer-3	200	10V	10A	100	5V	20A

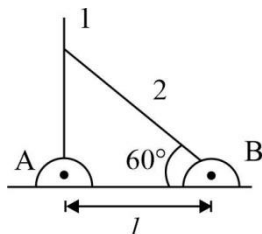
11. The primary coil of a transformer has 100 turns and is connected to a 120V AC source. How many turns are in the secondary coil if the voltage across it is 2400 V ?

- (A) 5 (B) 50 (C) 200 (D) 2000

12. A transformer with 40 turns in its primary coil is connected to a 120 V AC source. If 20W of power is supplied to the primary coil, how much power is developed in the secondary coil?
 (A) 10W (B) 20W (C) 80W (D) 160W

Paragraph for Question 13 - 14

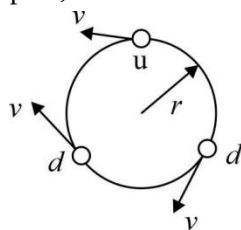
Two rods 1 and 2 are released from rest as shown in figure. Given: $l_1 = 4l$, $m_1 = 2m$, $l_2 = 2l$ and $m_2 = m$. There is no friction between the two rods. If α be the angular acceleration of rod 1 just after the rods are released. Then:



13. What is the normal reaction between the two rods at this instant?
 (A) $16\sqrt{3}ml\alpha$ (B) $\frac{4ml\alpha}{\sqrt{3}}$ (C) $\frac{32ml\alpha}{3\sqrt{3}}$ (D) $12\sqrt{3}ml\alpha$
14. What is the horizontal force on rod 1 by hinge A at this instant?
 (A) $\left(\frac{32-12\sqrt{3}}{3\sqrt{3}}\right)ml\alpha$ (B) $\left(\frac{16-2\sqrt{3}}{\sqrt{3}}\right)ml\alpha$
 (C) $(14+2\sqrt{3})ml\alpha$ (D) $\sqrt{3}ml\alpha$

Paragraph for Question 15 - 16

The neutron is a particle with zero charge still it has nonzero magnetic moment with z-component $9.66 \times 10^{-27} \text{ A} \cdot \text{m}^2$. This can be explained by the internal structure of the neutron. The evidence indicates that a neutron is composed of three fundamental particles called quarks: an “up” (u) quark, of charge $+\frac{2e}{3}$, and two “down” (d) quarks, each of charge $-\frac{e}{3}$. The combinations of the three quarks produces a net charge of $\frac{2e}{3} - \frac{e}{3} - \frac{e}{3} = 0$. If the quarks are in motion, they can produce a nonzero magnetic moment. As a very simple model, suppose the u quark moves in a counterclockwise circular path and the d quarks move in a clockwise circular path, all of the same radius r and all with the same speed v (see figure).



15. The current due to the circular motion of the u quark is :
 (A) $\frac{ev}{6\pi r}$ (B) $\frac{ev}{3\pi r}$ (C) $\frac{ev}{\pi r}$ (D) $\frac{2ev}{\pi r}$
16. Determine the magnitude of the magnetic moment due to the circular u quark.
 (A) $\frac{evr}{3}$ (B) $\frac{2evr}{3}$ (C) $\frac{4evr}{3}$ (D) evr

SECTION-3

This section contains 4 Multiple choice questions. Each question has matching lists. The codes for the lists have choices (A), (B), (C) & (D) out of which ONLY ONE is correct.

17. A ray of light falls normally on an equilateral prism of refractive index $\mu = \sqrt{3}$. Match the following lists :

List-I		List-II	
P	Angle of incidence while entering	1	0°
Q	Angle of refraction while entering	2	90°
R	Angle of emergence	3	60°
S	Angle of incidence inside prism before emerging	4	None

Codes:

	P	Q	R	S		P	Q	R	S
(A)	1	1	4	3	(B)	3	2	1	4
(C)	4	2	1	3	(D)	4	1	2	3

18. Match the following lists :

List-I		List-II	
P	Boltzmann constant	1	$[ML^2T^{-1}]$
Q	Coefficient of viscosity	2	$[ML^{-1}T^{-1}]$
R	Planck constant	3	$[MLT^{-3}K^{-1}]$
S	Thermal conductivity	4	$[ML^2T^{-2}K^{-1}]$

Codes:

	P	Q	R	S		P	Q	R	S
(A)	3	1	2	4	(B)	3	2	1	4
(C)	4	2	1	3	(D)	4	1	2	3

19. Standing waves are set up on strings fixed at both ends. In each case in column -I, ℓ is length of the string, c is the wave speed and f is the wave frequency. Each choice in column -II lists respectively the number of nodes and antinodes formed on the string. Number of nodes includes the ones formed at the two ends.

Column -I		Column -II	
P	$\ell = 0.5 \text{ m}, c = 15 \text{ m/s}, f = 30 \text{ Hz}$	1	2, 1
Q	$\ell = 0.4 \text{ m}, c = 20 \text{ m/s}, f = 50 \text{ Hz}$	2	3, 2
R	$\ell = 0.6 \text{ m}, c = 15 \text{ m/s}, f = 37.5 \text{ Hz}$	3	2, 3
S	$\ell = 0.5 \text{ m}, c = 25 \text{ m/s}, f = 100 \text{ Hz}$	4	3, 4
		5	4, 3
		6	5, 4

	P	Q	R	S		P	Q	R	S
(A)	1	1	4	2	(B)	3	3	4	6
(C)	2	2	5	6	(D)	2	5	6	6

20. Match the following lists.

List-I		List-II	
P	Sun	1	Nuclear fission
Q	Nuclear reactors	2	Nuclear fusion
R	Total binding energy in a process is increased	3	Energy is released
S	Total binding energy in a process is decreased	4	Energy is absorbed

Codes:

	P	Q	R	S		P	Q	R	S
(A)	2, 3	1, 3	1, 2, 3	4	(B)	3	2	1	4
(C)	4	2	1	3	(D)	4	1	2	3

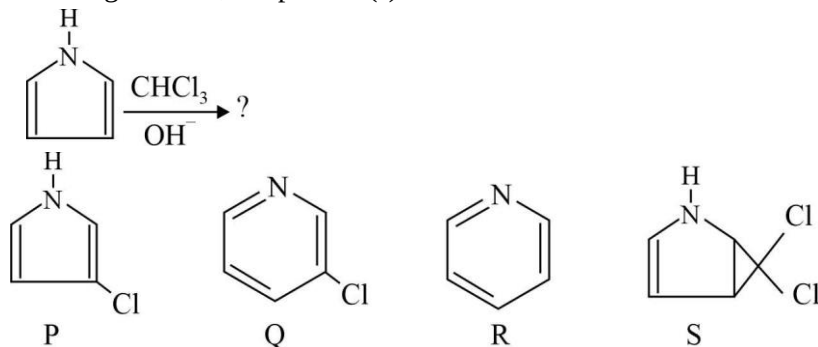
PART II : CHEMISTRY**68 MARKS****SECTION-1**

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. The K_{sp} of PbI_2 is 1.4×10^{-8} at 298K. The solubility of PbI_2 (in mol/L) in a $0.01M Pb(NO_3)_2$ solution is :

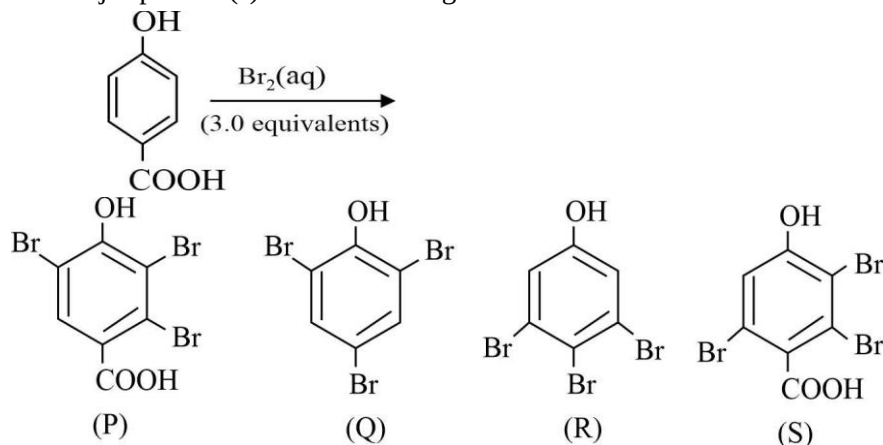
(A) 1.9×10^{-4} (B) 5.9×10^{-4} (C) 4×10^{-2} (D) 1.8×10^{-4}

2. In the following reaction, the product(s) formed is/are :



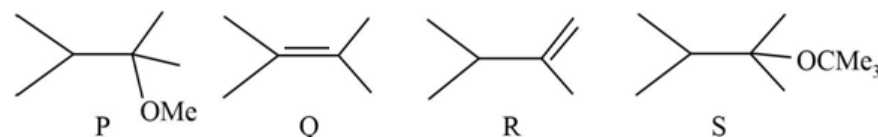
(A) P (major) (B) Q (major) (C) R (minor) (D) S (minor)

3. The major product(s) of the following reaction is/are :



(A) P (B) Q (C) R (D) S

4. After completion of reactions (I & II), the major organic compound(s) in the reaction mixture is/are :



(A) Reaction I : P and Reaction II : Q (B) Reaction I : Q and Reaction II : R
(C) Reaction I : P and Reaction II : S (D) Reaction I : R and Reaction II : S

5. The correct statements(s) about O_3 is/are :
- (A) Ozone is a powerful oxidising agent as compared to O_2
 (B) Ozone reacts with KOH and gives an orange coloured solid KO_3
 (C) There is a decrease in volume when ozone decomposed to form O_2
 (D) The decomposition of O_3 to O_2 is exothermic
6. In the nuclear reaction ${}^{68}_{32}\text{Ge} + {}^0_{-1}\text{e} \longrightarrow {}^A_Z\text{Y} + \text{X-rays} + \nu$
- (A) $Z = 31, A = 68$ (B) $Z = 32, A = 72$
 (C) $Z = 33, A = 67$ (D) $Z = 30, A = 72$
7. In which of the following metallurgy, no reducing agent is required from outside?
- (A) Mercury from cinnabar (B) Zinc from zinc blende
 (C) Iron from haematite (D) Aluminium from Bauxite
8. At a certain temperature (T_1), Some hydrazine (N_2H_4) gas is placed in a sealed flask where the following equilibrium is established.
- $$3N_2H_4(g) \xrightleftharpoons{K_1} 4NH_3(g) + N_2(g) (\Delta H^\circ > 0)$$
- If the temperature is increased to T_2 , some ammonia gas also decomposes as
- $$2NH_3(g) \xrightleftharpoons{K_2} N_2(g) + 3H_2(g) (\Delta H^\circ > 0)$$
- What would be observed as temperature is increased from T_1 to T_2 ?
- (A) Some more hydrazine would decompose
 (B) If $K_1 < K_2$, concentration of NH_3 would decrease
 (C) Partial pressure of N_2 would be doubled
 (D) All of these would be observed

SECTION-2

This section consists of **FOUR (04) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question 9 - 10

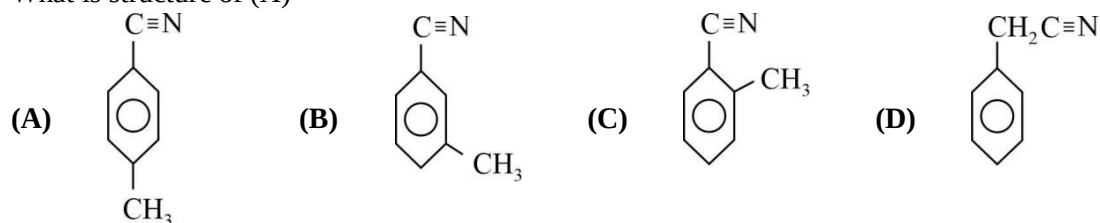
Analysis of II group of basic radicals involve precipitation of the metal ion as insoluble metal sulphide. The group reagent used is dilute HCl and saturated H_2S .

9. Which of following pair of metal ions can't be separated using the group reagent of II group of basic radicals.
- (A) Pb^{+2} and Cu^{+2} (B) Hg^{+2} and Mn^{+2}
 (C) Cu^{+2} and Ni^{+2} (D) Bi^{+3} and Fe^{+3}
10. Which of following metal sulphide of II group is insoluble in yellow $(NH_4)_2S$?
- (A) As_2S_3 (B) SnS_2 (C) SnS (D) HgS

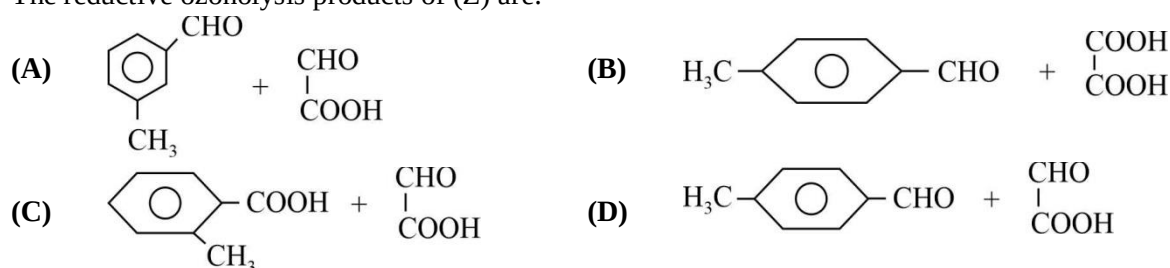
Paragraph for Question 11 - 12

Compound (X) (C_8H_7N) on treatment with cold $SnCl_2$ / dil. HCl (aqueous), produces Y (C_8H_8O). (Y) on heating with sodium acetate in acetic anhydride followed by acidification forms (Z) ($C_{10}H_{10}O_2$). All these compounds (X, Y, Z) on oxidation under severe conditions form a diacid which gives only one mononitro isomer in nitrating mixture.

11. What is structure of (X)

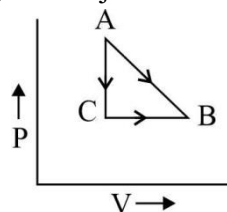


12. The reductive ozonolysis products of (Z) are:



Paragraph for Question 13 - 14

A fixed mass 'm' of a gas is subjected to transformation of state 'A' to state 'B' as shown



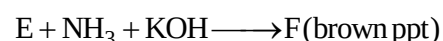
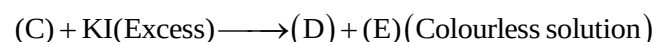
13. If length $AC = CB$ then select correct statement regarding the P-V diagram

- (A) $\Delta H = q_p$ for path $A \rightarrow C$ (B) $\Delta S_{A \rightarrow B} = \Delta S_{A \rightarrow C \rightarrow B}$
(C) $W_{A \rightarrow B} = W_{A \rightarrow C \rightarrow B}$ (D) $W = C + ve$ for path $A \rightarrow C \rightarrow B$

14. The pair of isochoric and isobaric process among the transformation of states is

- (A) A to C and C to B (B) C to B and A to B
(C) A to C and A to B (D) None of these

Paragraph for Question 15 - 16



15. Compounds (A) and (B) are respectively

- (A) $AgNO_3$ and $AgCl$ (B) $Pb(NO_3)_2$ and $PbCl_2$
(C) $Hg_2(NO_3)_2$ and Hg_2Cl_2 (D) $Cu_2(NO_3)_2$ and Cu_2Cl_2

16. When compound (A) reacts with Na_2CrO_4 solution, the colour of the compound formed is :

- (A) Black (B) Red (C) Yellow (D) White

SECTION-3

This section contains 4 Multiple choice questions. Each question has matching lists. The codes for the lists have choices (A), (B), (C) & (D) out of which ONLY ONE is correct.

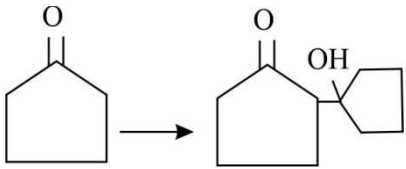
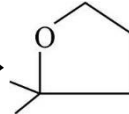
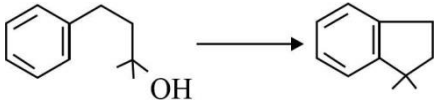
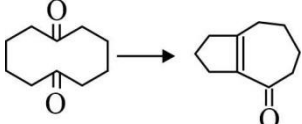
17. The unbalanced chemical reactions given in List-I show missing product which are provided in List-II. Match list-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
P	$\text{NaBr} + \text{MnO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{MnSO}_4 + \text{H}_2\text{O} + \text{other product}$	1	HIO_3
Q	$\text{I}_2 + \text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{other product}$	2	I_2
R	$\text{CaOCl}_2 + \text{CH}_3\text{COOH} + \text{KI} \rightarrow (\text{CH}_3\text{COO})_2\text{Ca} + \text{KCl} + \text{H}_2\text{O} + \text{other product}$	3	CrO_2Cl_2
S	$\text{NaCl} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{KHSO}_4 + \text{H}_2\text{O} + \text{other product}$	4	Br_2

Code:

	P	Q	R	S		P	Q	R	S
(A)	4	1	3	2	(B)	3	2	1	4
(C)	1	2	4	3	(D)	4	1	2	3

18. Match the chemical conversions in List-I with the appropriate reagents in List-II and select the correct answer using the code given below the lists.

List-I		List-II	
P		1	H_2SO_4
Q	$\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{Cl} \longrightarrow$ 	2	dil KOH/ Δ
R		3	aq. NaOH
S		4	CH_3MgI

Code:

	P	Q	R	S		P	Q	R	S
(A)	3	4	1	2	(B)	3	2	1	4
(C)	1	2	4	3	(D)	2	1	3	4

19. An aqueous solution of X is added slowly to an aqueous solution of Y as shown in List-I. The variation in conductivity of these reactions in List-II. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
P	Ph COOH + Ph NH ₂ X Y	1	Conductivity decreases and then increases
Q	NaCl (aq) + AgNO ₃ (aq) X Y	2	Conductivity decreases and then does not change much
R	KOH + HCl X Y	3	Conductivity increases and does not change much
S	CH ₃ COOH + NaOH X Y	4	Conductivity does not change much and then increases

Code:

	P	Q	R	S		P	Q	R	S
(A)	4	1	2	3	(B)	2	1	4	3
(C)	3	4	1	2	(D)	3	2	1	4

20. The standard reduction potential data at 25°C is given below :

$$E^\circ(\text{MnO}_4^- / \text{Mn}^{2+}) = 1.51\text{V}$$

$$E^\circ(\text{MnO}_2 / \text{Mn}^{2+}) = 1.23\text{V}$$

$$E^\circ(\text{ClO}_3^- / \text{ClO}^-) = 0.54\text{V}$$

$$E^\circ(\text{ClO}^- / \text{Cl}^-) = 0.76\text{V}$$

$$E^\circ(\text{Au}^{3+} / \text{Au}) = 1.5\text{V}$$

$$E^\circ(\text{Au}^{3+} / \text{Au}^+) = 1.4\text{V}$$

$$E^\circ(\text{Cr}^{3+} / \text{Cr}) = -0.74\text{V}$$

$$E^\circ(\text{Cr}^{2+} / \text{Cr}) = -0.91\text{V}$$

Match E° of the redox pair in List-I with the values given in List-II and Select the correct answer using the code given below the list.

List-I		List-II	
P	$E^\circ(\text{MnO}_4^- / \text{MnO}_2)$	1	0.61V
Q	$E^\circ(\text{ClO}_3^- / \text{Cl}^-)$	2	-0.4 V
R	$E^\circ(\text{Au}^+ / \text{Au})$	3	1.7 V
S	$E^\circ(\text{Cr}^{3+} / \text{Cr}^{2+})$	4	1.69 V

Code:

	P	Q	R	S		P	Q	R	S
(A)	1	2	3	4	(B)	4	1	3	2
(C)	3	4	2	1	(D)	4	3	1	2

PART III : MATHEMATICS**68 MARKS****SECTION-1**

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

- If $a, b \in \mathbb{R}^+$ then $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{(k+an)(k+bn)}$ is equal to :

(A) $\frac{1}{a-b} \ln \frac{b(b+1)}{a(a+1)}$ if $a \neq b$ (B) $\frac{1}{a-b} \ln \frac{a(b+1)}{b(a+1)}$ if $a \neq b$

(C) non existent if $a = b$ (D) $\frac{1}{a(1+a)}$ if $a = b$
- The equation to the circle which passes through the origin and cut off equal chords of length "a" from the straight lines $y = x$ and $y = -x$ can be :

(A) $x^2 + y^2 + \sqrt{2}ax = 0$ (B) $x^2 + y^2 - \sqrt{2}ax = 0$

(C) $x^2 + y^2 + \sqrt{a}x = 0$ (D) None of these
- If the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = q$, $\vec{r} \cdot (\hat{i} + 2a\hat{j} + \hat{k}) = q$ and $\vec{r} \cdot (a\hat{i} + a^2\hat{j} + \hat{k}) = q$ intersect in a line, then the value of a is :

(A) 1 (B) 1/2 (C) 2 (D) 0
- If $\sin A, \sin B$ are the roots of the equation $c^2x^2 - c(a+b)x + ab = 0$ where A, B, C are the angles and a, b, c are the sides of a $\triangle ABC$ opposite to A, B, C respectively, then the triangle :

(A) is obtuse angled (B) is acute angled

(C) is right angled (D) satisfies $c(\sin A + \cos A) = a + b$
- Let $z \in \mathbb{C}$ and if $A = \{z : \arg(z) = \pi/4\}$ and $B = \{z : \arg(z - 3 - 3i) = 2\pi/3\}$. Then $n(A \cap B)$ is not greater than :

(A) 1 (B) 2 (C) 3 (D) 0
- The solution set of $|x-1|^{\log_2(4-x)} < |x-1|^{\log_2(1+x)}$ is $((0, a) \cup (b, c)) - \{d\}$ then :

(A) $a = 3$ (B) $b = 2$ (C) $c = 4$ (D) $abcd = 12$
- If A and B are square matrices of order 2 such that $A + \text{adj}(B^T) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $A^T - \text{adj}(B) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then :

(A) B is symmetric matrix (B) $A^n = nA \forall n \in \mathbb{N}$

(C) $|A + A^2 + A^3 + A^4 + A^5| = 0$ (D) $|B + B^2 + B^3 + B^4 + B^5| = 0$
- Greatest value of $f(x) = \frac{1}{2ax - x^2 - 5a^2}$ in $[-3, 5]$ is :

(A) $f(5)$ if $a = 1$ (B) $f(-3)$ if $a = 1$ (C) $f(5)$ if $a < 1$ (D) $f(-3)$ if $a > 1$

SECTION-2

This section consists of **FOUR (04) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question 9 - 10

Let $f(x)$ be a non-positive continuous function and $F(x) = \int_0^x f(t) dt \forall x \geq 0$ and $f(x) \geq cF(x)$ where $c > 0$

and let $g : [0, \infty) \rightarrow R$ be a function such that $\frac{dg(x)}{dx} < g(x) \forall x > 0$ and $g(0) = 0$.

9. The total number of root(s) of the equation $f(x) = g(x)$ is/are :
(A) ∞ **(B)** 1 **(C)** 2 **(D)** 0
10. The number of solution(s) of the equation $|x^2 + x - 6| = f(x) + g(x)$ is/are :
(A) 2 **(B)** 1 **(C)** 0 **(D)** 3

Paragraph for Question 11 - 12

Straight line $x + y = 18$ and common tangents of the curve $S_1 = x^2 + y^2 - 28x + 160 = 0$ and $S_2 = \frac{x^2}{64} + \frac{y^2}{36} - 1 = 0$ form a triangle ABC in the first quadrant and $S_3 = 0$ is circumcircle of $\triangle ABC$.

11. Equation of S_3 is :
(A) $x^2 + y^2 - 16x - 20y + 156 = 0$ **(B)** $x^2 + y^2 - 10x - 8y + 156 = 0$
(C) $x^2 + y^2 - 8x - 10y + 156 = 0$ **(D)** $x^2 + y^2 - 20x - 16y + 156 = 0$
12. Length of common chord of circle S_1 and S_3 :
(A) is $\frac{3\sqrt{5}}{10}$ **(B)** is $\frac{3\sqrt{5}}{20}$ **(C)** is $\frac{7\sqrt{5}}{10}$ **(D)** does not exist

Paragraph for Question 13 - 14

Suppose z and w be two complex numbers such that $|z| \leq 1, |w| \leq 1$ and $|z + iw| = 2$. Answer the following questions.

13. Which of the following is true about $|z|$ and $|w|$?
(A) $|z| = |w| = \frac{1}{2}$ **(B)** $|z| = \frac{1}{2}, |w| = \frac{3}{4}$ **(C)** $|z| = |w| = \frac{3}{4}$ **(D)** $|z| = |w| = 1$
14. Which of the following is true for z and w ?
(A) $\operatorname{Re}(z) = \operatorname{Re}(w)$ **(B)** $\operatorname{Im}(z) = \operatorname{Im}(w)$
(C) $\operatorname{Re}(z) = \operatorname{Im}(w)$ **(D)** $\operatorname{Im}(z) = \operatorname{Re}(w)$

Paragraph for Question 15 - 16

There are 10 numbered boxes $B_1, B_2, \dots, B_{10}$ each box has total 11 balls such that i^{th} box contains i white balls and $(11-i)$ red balls. E denotes the event of selecting a white ball and A_i denote the event of selecting i^{th} box. If selection of each box is equally likely, then

15. $P(E)$ is :
 (A) $3/4$ (B) $1/5$ (C) $1/2$ (D) None of these
16. Value of $P(A_6 / E)$ is :
 (A) $2/55$ (B) $3/55$ (C) $4/55$ (D) $6/55$

SECTION-3

This section contains 4 Multiple choice questions. Each questions has matching lists. The codes for the lists have choices (A), (B), (C) & (D) out of which ONLY ONE is correct.

17. Match the column :

List-I		List-II	
P	The number of solutions of $ \cot x = \cot x + \operatorname{cosec} x, 0 \leq x \leq 2\pi$, is	1	1
Q	If m and n ($n > m$) are positive integers, the number of solutions of equation $n \sin x = m \cos x $ in $[0, 2\pi]$ is	2	2
R	If $[\sin x] + \left[\frac{x}{2\pi}\right] + \left[\frac{2x}{5\pi}\right] = \frac{9x}{10\pi}$, then the number of solutions in the interval $(30, 40)$ is (where $[.]$ represents greatest integer function)	3	3
S	The value of x in $(0, \pi/4)$ satisfying the equation, $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ is $\frac{\pi}{k}$ then the divisor of k is	4	4

Codes:

	P	Q	R	S		P	Q	R	S
(A)	1	4	1	1, 2, 3, 4	(B)	1	2	4	1, 3, 4
(C)	2	1	4	1, 2, 3	(D)	2	4	1	1, 2, 3, 4

18. Match the following :

List-I		List-II	
P	Common normals to the parabola $y^2 = 4ax$ and $x^2 = 4ay$ is/are	1	$x = a$
Q	The locus of point P if tangents from P to the parabola $y^2 = 4ax$ intersect the co-ordinate axes in concyclic points	2	$x + y = 3a$
R	The locus of P , if tangents from it to the parabolas $y^2 = 4a(x+a)$ and $y^2 = 8a(x+2a)$ are perpendicular	3	$x + 3a = 0$
S	The chord of contact of a point P w.r.t. the parabola $y^2 + 4ax = 0$ subtends right angle at the vertex. The locus of P is	4	$x = 4a$

Codes:

	P	Q	R	S		P	Q	R	S
(A)	1	2	3	4	(B)	2	1	3	4
(C)	1	2	4	3	(D)	1	4	2	3

19. Match the items of column I with that of column II:

List-I		List-II	
P	$[2\vec{a} \ 3\vec{b} \ 5\vec{c}]$	1	$[\vec{a} \ \vec{b} \ \vec{c}]^4$
Q	$[\vec{a}+\vec{b} \ \vec{b}+\vec{c} \ \vec{c}+\vec{a}]$	2	$30[\vec{a} \ \vec{b} \ \vec{c}]$
R	$[(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) \ (\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b}) \ (\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c})]$	3	$2[\vec{a} \ \vec{b} \ \vec{c}]$
S	$[\vec{b} \times \vec{c} \ \vec{c} \times \vec{a} \ \vec{a} \times \vec{b}]$	4	$[\vec{a} \ \vec{b} \ \vec{c}]^2$

Codes:

	P	Q	R	S		P	Q	R	S
(A)	2	3	1	4	(B)	1	2	3	4
(C)	2	1	4	3	(D)	1	2	4	3

20. Match the following column :

List-I		List-II	
P	The coordinates of a point on the line $x=4y+5, z=3y-6$ at distance 3 from the point $(5, 3, -6)$ is/are	1	$(-1, -2, 0)$
Q	The plane containing the line $\frac{x-2}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and parallel to $\hat{i} + 4\hat{j} + 7\hat{k}$ has the point	2	$(5, 0, -6)$
R	A line passes through two points $A(2, -3, -1)$ and $B(8, -1, 2)$. The coordinates of a point of this line farther from the origin and at a distance of 14 units from A is/are	3	$(2, 5, 7)$
S	The coordinates of the foot of the perpendicular from the point $(3, -1, 11)$ on the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is/are	4	$(14, 1, 5)$

Codes:

	P	Q	R	S		P	Q	R	S
(A)	1	2	3	4	(B)	2	1	3	4
(C)	2	1	4	3	(D)	1	2	4	3